# Exam 3 Review Session Notes

Tuesday, April 8, 2025 11:56 AM

**Question:** Silver has the FCC crystal structure with a unit cell parameter $a_0 = 4.09$ Å. Each atom is coordinated to 12 neighbors. Given that the molar heat of sublimation for Ag is 284 kJ/mol, estimate the surface energy (in $J/cm^2$) of the Ag (100) surface.

Ag (FCC) (100)

$FCC_{(100)} = N/a^2 = 2/(4.09e\text{-}10\ m)^2$

Surface Energy ($\gamma$):

$$\gamma_{Ag(100)} = \frac{2}{(4.09e\text{-}10m)^2} \cdot \frac{284000}{3 \times 6.02e23}$$

$$\gamma_{Ag(100)} = 1.88\ J/m^2 = 1.88e\text{-}4\ J/cm^2$$

* What if given the radius of an atom instead of $a$

- FCC → $a = 4r/\sqrt{2}$
- BCC → $a = 4r/\sqrt{3}$
- SC → $a = 2r$
- Diamond → $a = 8r/\sqrt{3}$

**Question:** Silicon is an intriguing potential anode material for Li ion batteries because it can store and release more than four Li atoms per Si atom. Quantifying the diffusivity of Li in Si is an important aspect for determining the charge/discharge kinetics of such a battery material. If the diffusivity of Li in Si is $4.0 \times 10^{-14}$ $cm^2/s$ at 300 K and $2.0 \times 10^{-11}$ $cm^2/s$ at 400 K, determine $\Delta G_{act}$ and $D_0$ for the diffusion of Li in Si.

$$\frac{D_1 = D_0 e^{-\Delta G_{act}/RT_1}}{D_2 = D_0 e^{-\Delta G_{act}/RT_2}} = e^{-\Delta G_{act}/R\,(1/T_1 - 1/T_2)}$$

- Solve for $\Delta G_{act}$:

$$\Delta G_{act} = R\,\frac{T_2 T_1}{T_1 - T_2} \ln(D_1/D_2) = 8.314\ \frac{300(400)}{300-400} \ln\left(\frac{4e\text{-}14}{2e\text{-}11}\right)$$

$\Delta G_{act} = 62$ kJ/mol

↳ plug back in to find $D_0$

$4e\text{-}14 = D_0 e^{-62\,kJ/mol/8.314(300)}$

$D_0 = 2.5e\text{-}3\ cm^2/s$

True or False:

True or False:

- Film thickness in an ALD Process is determined by the time of the process
  - False - determined by # of cycles

**Question:** In the atomic layer deposition of alumina ($Al_2O_3$), the average deposition rate is approximately 0.5 ML (ML = monolayer) per cycle. A single cycle consists of two purge steps and two reaction steps. Even with careful attention to valve, chamber, and gas flow design, typically at least a few seconds is required to purge the chamber, introduce the precursor gas, and then ensure that all surfaces undergo complete reaction for each step. For the purposes of this question, assume that one complete cycle takes 10 s. Based on this assumption, determine the time it takes to deposit a 1-μm-thick film of $Al_2O_3$ by ALD. Assume that 1 ML of $Al_2O_3$ is 4 Åthick.

$$dx/dt = \frac{0.5\,ML}{10\,seconds} \rightarrow \frac{0.5\,ML \cdot 4\,Å/ML}{10\,seconds} = \frac{2\,Å}{10\,seconds} = 0.2\,Å/s = 72\,nm/h$$

For 1μm (1e3 nm):

$$\frac{1e3\,nm}{72\,nm/h} = 14\text{ hours}$$

- Takes a really long time, whats an alternative?
  - CVD → really expensive → no
  - Physical Vapor Deposition (PVD) → quicker and cheaper ✓

**Question**

A vacuum reactor is used to synthesize $WO_3$ nanoparticles for use in an electrochromic window device. To create the $WO_3$ nanoparticles, a tungsten (W) metal filament 1.0 mm in diameter and 10 cm long is heated to 2,200°C in a vacuum with a trace pressure of oxygen gas. Assuming all of the W evaporating from the surface of the filament is rapidly and completely oxidized to $WO_3$, determine the maximum rate of production of $WO_3$ from this reactor (g/s). The equilibrium vapor pressure of W at 2,200°C is approximately $10^{-10}$ atm.

$$2W + 3O_2 = 2WO_3$$

Molecular Flux Reaction where:

$\alpha = 1$ $\quad M_W = 184.3\ g/mol$

$P_W^{Eq} = 10^{-10}\ atm$ $\quad T = 2473\ K$

$$J_{W,wire} = \frac{\alpha P_W^{Eq}}{\sqrt{2\pi M_W RT}} = \frac{10^{-10}\ atm\ (101325\ Pa/atm)}{\sqrt{2\cdot\pi\cdot 0.1843\ kg/mol \cdot 2473\ K \cdot 8.314\ J/mol\cdot K}}$$

$\hookrightarrow J = kg \cdot m^2/s^2$

$$J_{W,wire} = 6.581\ e^{-8}\ mol/m^2\cdot s$$

Rate of $WO_3$ Production $= A J_W \cdot M_{WO_3}$

$= 3.142\ e\text{-}4\ m^2 \times 6.581\ e\text{-}8\ mol/m^2\cdot s \times 23.8\ g/mol = 4.8\ e\text{-}9\ g/s$

True or False:

- In the passive oxidation of silicon

$$Si_{(s)} + O_{2(g)} \rightarrow SiO_{2(s)}$$

oxygen diffusion controls the oxidation rate

False

Si diffusion in $SiO_2$ surface

- slowest process, thus one controls the rate

Questions 1 and 3 on Homework 9 are very important